

Why frequency is important

Consider the following example. A whistle blows at about 45dB, while a thunder is at 100dB. But, a thunder has frequency of only 50Hz, whereas the whistle has a frequency of 1000Hz or more. Decibel level is not enough to understand the quality of sound; frequency is equally important. Hence, an SLM that has better response over a large frequency range is considered more accurate.

Leq (equivalent continuous sound level). In industries, noise-measurement time-averaged value (in decibels) is used instead of sound exposure level. Leq is the average sound pressure over a period of time. Unlike numerical values, decibel values cannot be averaged directly because these are logarithmic values. Hence, these cannot be added or subtracted directly.

SLMs automatically calculate Leq during and after the measurement has finished. These sample the noise level 16 times per second and then convert dB readings back into sound pressure levels, add these up, divide by the number of samples and return Leq back in dB form. SLMs that integrate such logarithmic values by taking samples at frequent intervals over a period of time are integrating SLMs.

Time weighting (fast/slow response time). Sound level measurements using any grade of SLM can be fast (125 milliseconds time constant) and slow (one second time constant). These weightings date back to the time when SLMs had analogue meters and defined the speed at which the meters moved. As these time constants were set in

SOUND LEVEL METERS

Make/Brand	Description
AMPROBE (SM-20A)	• IEC 651 type 2, ANSI 1.4 type 2 • Dynamic range: 50dB • Time weighting: Fast (125 milliseconds) and slow (1 second) • Frequency range: 31.5Hz to 8kHz • Frequency weighting: A and C
B&K Precision (735)	• IEC 651 type 2 • Accuracy: ± 1.5dB • Auto ranging: 30dB~130dB • Resolution: 0.1dB • 32,000 records data logger • Frequency weighting: A and C
Cirrus Optimus (CR-17x Series)	• IEC61672-1 class 1 and ANSI S1.4 type 2 • Frequency weightings: Simultaneous A, C and Z • Time weightings: Simultaneous F, S and I • Measurement range: Single 120dB • 4GB memory capable of storing over 10,000 measurements • Voice tag audio recording • Acoustic fingerprint audio recording
HTC Instruments (SL-1352)	• IEC61672-1 class 2 and ANSI S1.4 type 2 • Frequency weighting: A and C • Time weighting: Fast (125 milliseconds) and slow (one second) • Data log/gate function and memory for 262,100 readings
HTC Instruments (SL-1350)	• IEC61672-1 class 2 and ANSI S1.4 type 2 • Frequency weighting: A and C • Measurement parameters: LP, Leq and LN • Manual storage of 30 sets of data
Lutron (SL-4033SD)	• IEC 61672 class 1 • Auto range: 30dB to 130dB • Frequency weighting: A and C • Time weighting: Fast/slow • Available for external calibration adjustment • Real-time SD memory card data logger
TES (TES-1150/ TES-1151)	• IEC 61672-1 class 2, ANSI S1.4 type 2 • Dynamic range: 65dB • Frequency weighting: A, C and Z • Time weighting: Fast, slow and impulse

IEC 61672-1:2013 is the international standard that specifies SLM functionalities and performances

standards, these have continued in digital meters, now being calculated or simulated before being displayed,

stored or used to calculate other acoustic parameters that need time weighting.

Following terms are used to express sound level, clearly indicating time weighting as well as frequency weighting: LAS (slow, A weighted sound level), LAF (fast, A weighted sound level), LCS (slow, C weighted sound level) and LCF (fast, C weighted sound level).

Slow response is most often used for listening to conservations/OSHA programmes and is an averaging of sound level over time.

Fast response is most commonly used for product noise-reduction efforts and can detect quick noise bursts such as fireworks.

Sound level data logger.

Tango and EXTECH HD600 data-logging SLMs measure, display and store sound-level measurement data. Up to 20,000 readings can be stored in the meter, each with date and time stamp. Readings can be recalled on the meter's display or can be downloaded to a PC.

Included software permits users to view, graph, print and export data to a spreadsheet. This capability can be applied to many applications such as airport-noise analysis, construction-zone monitoring, traffic intersections, concert venues, environmental impact studies and others.

Noise dosimeter. These are ideal for industries where shifts are involved, such as construction (building sites), entertainment (orchestras) and emergency services (police forces). Rather than having a health and safety officer follow or stay with workers throughout assessment, each employee can be left to his or her own device, while